HK IMO Training 2020-2021
Problem Set 4 Suggested Solutions

Problem 13 (Austrian Regional Competition 2016 Problem 2). Let a, b, c and d be real numbers with $a^2 + b^2 + c^2 + d^2 = 4$. Prove the inequality

$$(a + 2)(b + 2) \geqslant cd,$$

and find all equality cases.

Solution. We have

$$\begin{aligned} 2[(a + 2)(b + 2) - cd] &= 2ab + 4a + 4b + 8 - 2cd \\ &= 2ab + 4a + 4b + 4 + (a^2 + b^2 + c^2 + d^2) - 2cd \\ &= (a + b + 2)^2 + (c - d)^2 \geqslant 0. \end{aligned}$$

So $(a + 2)(b + 2) \geqslant cd$. Equality holds when $a + b + 2 = c - d = 0$. Together with $a^2 + b^2 + c^2 + d^2 = 4$, this means $(a, b, c, d) = (a, -a - 2, \pm\sqrt{-a^2 - 2a}, \pm\sqrt{-a^2 - 2a})$ where $-2 \leqslant a \leqslant 0$. □

Problem 14. Let $c > 1$ be a real number. Let $\{x_n\}$ be a sequence of real numbers such that $x_n > 1$ and

$$x_1 + x_2 + \cdots + x_n < cx_{n+1}$$

for all positive integer n. Prove that there exists a real number $a > 1$ such that $x_n > a^n$ for any positive integer n.

Solution. We choose any $a > 1$ such that $a < c^{-1}$ and $a < x_1$. We prove the result by induction. The base case $x_1 > a$ holds by assumption. Assume $x_n > a^n$ for $n = 1, 2, \ldots, k$. Then

$$x_{k+1} > \frac{1}{c}(x_1 + x_2 + \cdots + x_k) > \frac{1}{c}(a + a^2 + \cdots + a^k) = \frac{1}{c} \cdot \frac{a^{k+1} - a}{a - 1} > \frac{a(a^{k+1} - a)}{a - 1}.$$

It remains to prove this is at least a^{k+1}. The inequality $\dfrac{a(a^{k+1} - a)}{a - 1} \geqslant a^{k+1}$ is equivalent to $\dfrac{a^k - 1}{a^k} \geqslant \dfrac{a - 1}{a}$. This holds since $f(x) = \dfrac{x - 1}{x} = 1 - \dfrac{1}{x}$ is increasing and $a^k \geqslant a$. So $x_{k+1} > a^{k+1}$, and we are done by induction. □

Problem 15 (Tuymaada Olympiad Senior 2020 Problem 2). Given positive real numbers $a_1, a_2, \ldots, a_n$, let

$$m = \min\left\{a_1 + \frac{1}{a_2}, a_2 + \frac{1}{a_3}, \ldots, a_{n-1} + \frac{1}{a_n}, a_n + \frac{1}{a_1}\right\}.$$

Prove the inequality

$$\sqrt[n]{a_1 a_2 \cdots a_n} + \frac{1}{\sqrt[n]{a_1 a_2 \cdots a_n}} \geqslant m.$$

Solution. We express all indices modulo n. There must be an index i such that $a_i \leqslant \sqrt[n]{a_1 a_2 \cdots a_n}$ since $\sqrt[n]{a_1 a_2 \cdots a_n}$ is the geometric mean. Let j be the smallest integer larger than i such that $a_j \geqslant \sqrt[n]{a_1 a_2 \cdots a_n}$, which must exist for the same reason. Then $a_{j-1} \leqslant \sqrt[n]{a_1 a_2 \cdots a_n} \leqslant a_j$ by definition. Now,

$$\sqrt[n]{a_1 a_2 \cdots a_n} + \frac{1}{\sqrt[n]{a_1 a_2 \cdots a_n}} \geqslant a_{j-1} + \frac{1}{a_j} \geqslant m.$$

□

Problem 16. Let a, b and c be positive real numbers. Find the minimum value of

$$\frac{(a+b)(b+c)(a+b+c)}{abc}.$$

Solution. Let $k = \dfrac{\sqrt{5}-1}{2}$ and let $d = \dfrac{b}{k}$. By the weighted AM-GM inequality, we have

$$a+b = a+kd \geqslant (1+k)(ad^k)^{\frac{1}{1+k}},$$
$$b+c = kd+c \geqslant (k+1)(d^k c)^{\frac{1}{k+1}},$$
$$a+b+c = a+kd+c \geqslant (1+k+1)(ad^k c)^{\frac{1}{1+k+1}}.$$

Since $k^2+k-1=0$, it is easy to verify $\dfrac{1}{k+1}+\dfrac{1}{k+2}=1$ and $\dfrac{2k}{k+1}+\dfrac{k}{k+2}=1$. Therefore, multiplying the above inequalities, we obtain

$$(a+b)(b+c)(a+b+c) \geqslant (k+1)^2(k+2)acd = \frac{(k+1)^2(k+2)}{k}abc = \frac{11+5\sqrt{5}}{2}abc.$$

Equality holds when $a=d=c$, i.e. when $ka=b=kc$. Therefore, the minimum value of $\dfrac{(a+b)(b+c)(a+b+c)}{abc}$ is $\dfrac{11+5\sqrt{5}}{2}$.

(Note that the value of k is chosen such that $\dfrac{1}{k+1}+\dfrac{1}{k+2}=\dfrac{2k}{k+1}+\dfrac{k}{k+2}=1$. Indeed, since the inequality is homogeneous, these hold if and only if one of these holds.) $\square$